

Business Problem Statement

The e-commerce industry generates large volumes of transactional data on a daily basis. While this data contains valuable information about customer behavior, product performance, sales trends, and business operations, organizations often struggle to convert raw data into actionable insights. Without proper analysis, businesses may miss opportunities to improve revenue, optimize product offerings, enhance customer satisfaction, and reduce operational inefficiencies.

This project focuses on analyzing an e-commerce sales dataset containing 34,500 customer orders collected between September 2023 and September 2025. The dataset includes information related to product categories, order values, customer demographics, payment methods, regions, delivery performance, and product returns.

The primary business challenge is to identify the key factors influencing sales performance and customer purchasing behavior. Specifically, the organization seeks to understand which product categories generate the highest revenue, which regions contribute most to sales, what payment methods customers prefer, how return rates impact business performance, and which customer segments provide the greatest value.

By applying data cleaning, exploratory data analysis, pivot table analysis, trend analysis, and dashboard visualization techniques in Microsoft Excel, this project aims to transform raw transactional data into meaningful business insights. The findings will help support data-driven decision-making, improve strategic planning, and identify opportunities for business growth and operational improvement.